Integrality of the alpha-triangle

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Status. Complete mathematical proof. The final published source has not been available for comparison; campaign eligibility remains pending, with zero new credit. Review was by the authoring agent only.

Task 1: Full solution

Govindarajan's Conjecture 3.1 in [1], printed page 16, asserts the integrality of the coefficients defined by

$$a_{m+r+1,r} = \frac{1}{(2m)!!} \sum_{z=0}^{2m} \alpha_{m,z} r^{2m-z}, \qquad (2m)!! = 2^m m!.$$

Here a Ferrers diagram in $\mathbb{Z}^r_{\geq 0}$ is a finite order ideal for the coordinatewise order. A strict diagram uses every coordinate. It therefore contains $\mu_r = \{0, e_1, \ldots, e_r\}$, and $a_{m+r+1,r}$ counts the diagrams obtained by adding m nodes to μ_r. These interpretations, and the orbit counting by reduced dimension used below, are prior results in [1], Sections 2.2–2.5.

Theorem 1. *For every $m \geq 0$, the polynomial $2^m m!\, a_{m+r+1,r}$ belongs to $\mathbb{Z}[r]$. Consequently every $\alpha_{m,z}$ is an integer. In fact this polynomial is a nonnegative integer linear combination of the falling factorials $(r)_k = r(r-1)\cdots(r-k+1)$.*

For a set S of added nodes, discard the coordinates that are zero on all of S. Its remaining number of coordinates is k, the reduced dimension. Thus $S \subset \mathbb{Z}^k_{\geq 0}$ uses every coordinate, $|S| = m$, every node of S has coordinate sum at least two, and $\mu_k \cup S$ is a Ferrers diagram. Let $\mathrm{Aut}(S) \leq \mathfrak{S}_k$ be its coordinate permutation group.

Lemma 2. *For every such reduced diagram, $|\mathrm{Aut}(S)|$ divides $2^m m!$.*

Proof. Let $T \subseteq S$ consist of the nodes with coordinate sum two. Construct a graph on the k coordinates: $2e_i \in T$ is a loop at i, and $e_i + e_j \in T$ is the undirected edge $\{i, j\}$. Every vertex occurs in this graph. Indeed, if $v \in S$ uses coordinate i, then either $v_i \geq 2$, so $2e_i \leq v$, or $v_i = 1$ and some $v_j > 0$, so $e_i + e_j \leq v$. The order-ideal condition supplies the stated quadratic node in each case. In particular $k \leq 2|T| \leq 2m$.

The group $G = \mathrm{Aut}(S)$ acts on the m added nodes. Write K for the kernel and H for the image of this action. Then $|G| = |K||H|$ and $|H| \mid m!$. Every member of K fixes each quadratic node individually. It therefore fixes every loop vertex and preserves each edge as an unordered pair.

A vertex on two distinct edges is fixed, as the unique intersection of their endpoint sets. Along an edge incident with a fixed vertex, the other endpoint is fixed as well. Hence all vertices of a connected component with a loop, or with at least three vertices, are fixed. The only components

admitting any remaining motion are components consisting of a single edge and its two endpoints, with no loops. If there are q such components, restriction to their endpoints embeds K into $(\mathbb{Z}/2\mathbb{Z})^q$. Notice that components cannot be permuted by K: every edge is fixed individually. Thus $|K| \mid 2^q$; and $q \le |T| \le m$. Combining the two divisibilities gives

$$|G| = |K||H| \;\mid\; 2^q m! \;\mid\; 2^m m!,$$

as required. Higher-degree nodes may further restrict K, which does not affect this argument. □

Proof of Theorem 1. For fixed m, let $\mathcal{R}_m$ contain one representative of each coordinate permutation orbit of reduced diagrams with m added nodes. It is finite: $k \le 2m$, and a coordinate value v_i in an added node is at most $m+1$, since all nodes $2e_i, \dots, v_i e_i$ must be present.

Fix $S \in \mathcal{R}_m$ with k active coordinates and $r \ge k$. There are $\binom{r}{k}$ ways to choose its coordinates in $\{1, \dots, r\}$ and $k!/|\operatorname{Aut}(S)|$ distinct coordinate labelings on a chosen set. The other coordinates contribute only their basis nodes in μ_r. Every diagram counted by $a_{m+r+1,r}$ arises uniquely in this fashion. Therefore

$$a_{m+r+1,r} = \sum_{S \in \mathcal{R}_m} \frac{(r)_{k(S)}}{|\operatorname{Aut}(S)|}. \tag{1}$$

This also holds for $0 \le r < k(S)$, since that summand vanishes. It is a polynomial identity: the right side is a fixed polynomial in r, and agrees with the defining count for every nonnegative integer r. Multiplication by $2^m m!$ and Lemma 2 give

$$2^m m!\, a_{m+r+1,r} = \sum_{S \in \mathcal{R}_m} \underbrace{\frac{2^m m!}{|\operatorname{Aut}(S)|}}_{\text{nonnegative integer}} (r)_{k(S)} \;\in \mathbb{Z}[r].$$

Its monomial coefficients are exactly the $\alpha_{m,z}$. For $m = 0$, the empty reduced diagram has $k = 0$, trivial automorphism group, and $(r)_0 = 1$; the polynomial is 1, as required. The prescribed zero entries outside the triangle are integers as well. □

Sharpness and an example. The only reduced diagram on $2m$ coordinates is a matching of m disjoint quadratic edges. To see this, the covering bound $k \le 2|T| \le 2m$ must be an equality: every added node is quadratic, no loop occurs, and no endpoints are shared. Its automorphism group has order $2^m m!$. Hence the leading coefficient of $a_{m+r+1,r}$ is $1/(2^m m!)$, and the normalization is the smallest positive integer that can make the whole polynomial integral. For $m = 1$, the loop and edge give

$$a_{r+2,r} = r + \frac{(r)_2}{2}, \qquad 2a_{r+2,r} = r^2 + r.$$

The same proof applies to any union of coordinate permutation orbits of reduced diagrams, chosen independently of the ambient dimension. In particular, it applies to the symmetric-box restrictions in Appendix A of [1].

Exact verification. The accompanying standard-library Python script independently enumerates all diagrams with $0 \le m \le 4$ in dimensions $1 \le r \le 8$. Their reduced counts match the source's Table 3. It checks the group, kernel, image, and orbit identities for all 87 reduced orbits in that range; compares 40 polynomial evaluations with direct counts; and checks 75 additional exact falling-factorial divisibilities in the source rows $m = 5, \dots, 9$. These finite checks supplement the proof; they are not used to infer the all-m assertion.

Task 2: The alpha–beta decomposition

Investigate whether decomposing reduced diagrams into connected components proves the finite binomial expansion of $\alpha_{m,z}$ proposed in Conjecture 3.2 of [1]. The present integrality proof does not establish that expansion or its asserted vanishing conditions.

Task 3: Denominators of restricted families

For families excluding isolated quadratic edges, determine the smallest normalizing integer. The proof isolates the possible kernel as a group of endpoint swaps, and may therefore permit sharper bounds than $2^m m!$.

Task 4: Weighted refinements

For integer-valued, coordinate-invariant statistics on reduced diagrams, study the resulting multivariate integral polynomials and identify natural combinatorial meanings for their falling-factorial coefficients.

Task 5: Computation by reduced orbits

Develop enumeration directly on reduced isomorphism types, keeping their automorphism orders. Formula (1) then recovers counts in every ambient dimension without enumerating all coordinate labelings.

References

[1] S. Govindarajan, *Notes on higher-dimensional partitions*, arXiv:1203.4419v1 (2012), 36 pages. Published in *J. Combin. Theory Ser. A* **120** (2013), 600–622, doi:10.1016/j.jcta.2012.11.005. The full preprint was read; the published version has not been read.